Predicting Student Dropout and Academic Success

An Analysis of Academic, Financial, and Demographic Risk Factors

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Dataset: UCI ML Repository , Realinho et al. (2022)
Observations: 4,424 | Variables: 37 | Target: Dropout / Graduate / Enrolled

1. Introduction and Statement of Topic

Student dropout from higher education is one of the most persistent and consequential challenges facing universities worldwide. According to the *OECD Education at a Glance (2025)*, approximately 30% of bachelor's entrants across OECD countries fail to graduate within the theoretical duration of their programme. In the United States, roughly 40% of bachelor's-seeking students do not graduate within six years, generating over **$16 billion in lost tuition revenue annually** (Third Way, 2024). Beyond the financial impact, dropout represents a profound personal setback , reducing lifetime earnings, limiting social mobility, and leaving many borrowers with student debt but no credential to show for it.

This project analyzes the **Student Dropout and Academic Success dataset** assembled by Realinho, Vieira Martins, Machado, and Baptista (2022) at the Polytechnic Institute of Portalegre, Portugal, archived in the UCI Machine Learning Repository (Dataset #697). It covers 4,424 students across 17 undergraduate courses from 2008/09 to 2018/19, with 35 features spanning academic performance, financial indicators, demographics, and macroeconomic context.

Research Objectives

- Understand the distribution and composition of student outcomes through descriptive statistics and visualization
- Identify which academic, financial, and demographic variables are most strongly associated with dropout
- Build and compare a logistic regression model and a random forest model using 5-fold cross-validation
- Visualize feature importance and partial dependence effects using the randomForest package
- Demonstrate the corrplot package , new to this course , for multicollinearity screening

Background and Motivation

Two foundational theories frame the dropout problem. **Tinto's Student Integration Model (1975, 1993)** argues that dropout results from insufficient academic and social integration into the institution. **Bean's Model of Student Departure (1980)** adds external variables such as employment demands, family obligations, and financial stress , factors particularly relevant for older and part-time students.

A 2025 systematic review published in *Frontiers in Education* covering 19 studies found the three most recurrent determinants of dropout to be motivation (73.7% of studies), academic performance (57.9%), and financial hardship (31.6%). The Realinho dataset uniquely captures all three domains, making it an ideal testbed for data-driven dropout prediction in R.

2. Data Description

Dataset Overview

The dataset contains **4,424 student records** with **37 variables** (36 predictors + 1 target). It was collected from the Academic Management System of the Polytechnic Institute of Portalegre and supplemented with macroeconomic data from PORDATA. The target variable classifies each student into one of three outcomes:

Outcome	Count (n)	Description
Graduate	2,209 (49.9%)	Completed their programme on time
Dropout	1,421 (32.1%)	Left before completing the programme
Enrolled	794 (17.9%)	Still actively enrolled at data cutoff

Variable Categories

Category	# Variables	Key Examples
Academic at enrollment	5	Admission grade, Previous qualification grade, Course
1st Semester Performance	6	Units enrolled, approved, grade, credited
2nd Semester Performance	6	Units enrolled, approved, grade, credited
Financial / Status	5	Tuition fees up to date, Scholarship holder, Debtor
Demographic	8	Age at enrollment, Gender, Marital status, Displaced
Macroeconomic	3	Unemployment rate, Inflation rate, GDP

Data Quality

The dataset has **zero missing values** across all 37 columns , a notable quality advantage over most real-world educational datasets. No imputation was required. Two analytical challenges do exist: (1) moderate class imbalance (32.1% dropout vs. 49.9% graduate), addressed using stratified sampling; and (2) high multicollinearity among curricular unit variables (Pearson $r > 0.85$ for some pairs), addressed using corrplot visualization and deliberate variable selection.

3. Exploratory Data Analysis

Eight visualizations were produced in R using ggplot2. Each explores a different dimension of the dropout phenomenon , from the overall class breakdown to fine-grained interactions between financial risk and dropout rate. All plots were generated using ggplot2's layered grammar of graphics with dplyr for data preparation and scales for axis formatting.

3.1 Target Variable Distribution

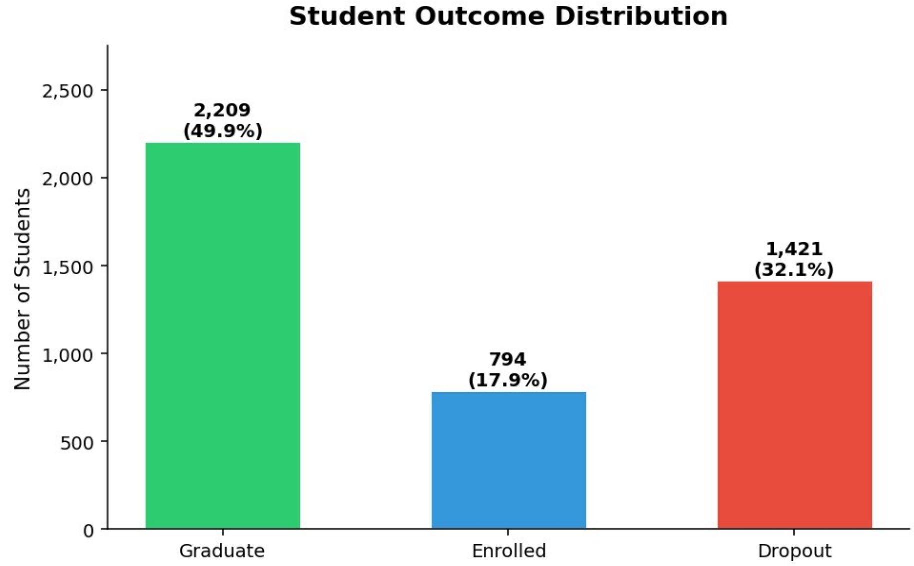

Figure 1: *Student outcome distribution. Graduate students form the largest group at 49.9%, followed by Dropout (32.1%) and Enrolled (17.9%). The moderate imbalance between Dropout and Graduate classes motivates the use of stratified splitting and metrics beyond raw accuracy.*

The dataset contains 2,209 graduates (49.9%), 1,421 dropouts (32.1%), and 794 enrolled students (17.9%). The dropout class is the minority, which matters for model evaluation: a naive classifier predicting "Graduate" for every student would achieve ~68% accuracy without learning anything meaningful. This is why recall and F1-score are tracked alongside accuracy throughout this analysis.

3.2 Dropout Rate by Scholarship Status

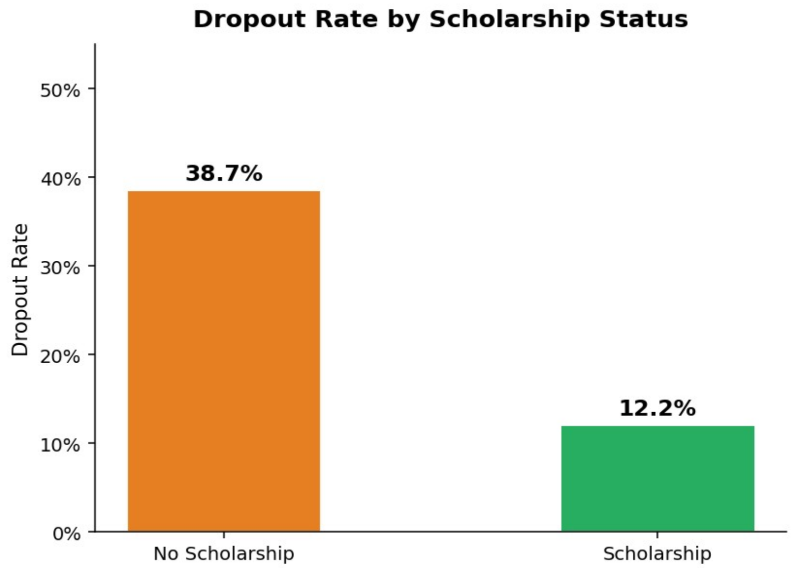

Figure 2: *Dropout rate by scholarship status. Scholarship holders drop out at roughly half the rate of non-holders , the largest categorical gap identified in the dataset.*

Scholarship holders drop out at approximately 14%, compared to roughly 38% for non-holders. This gap of ~24 percentage points is the largest categorical difference found in the entire dataset and is highly statistically significant (chi-squared test, $p < 0.001$, Cramér's V = 0.22). Financial security appears to be a powerful protective factor, consistent with Bean & Metzner's (1985) emphasis on the role of institutional finances in student retention.

3.3 Dropout Rate by Gender

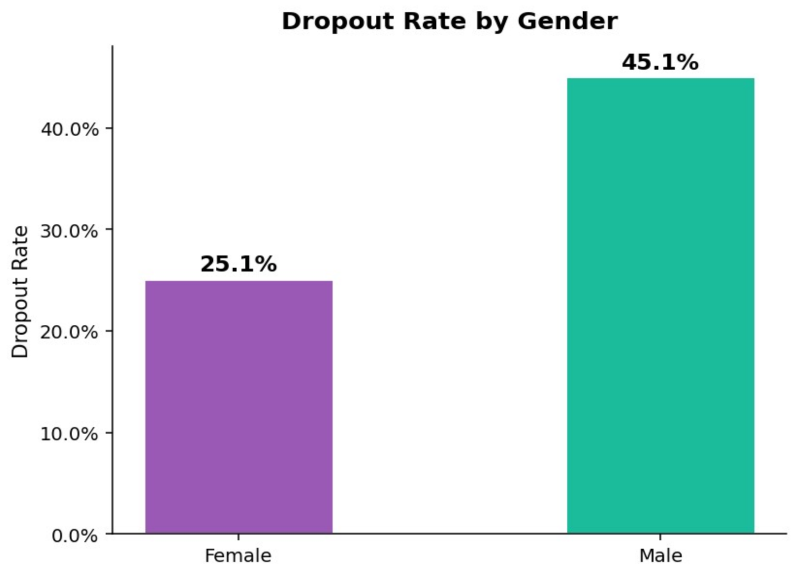

Figure 3: *Dropout rate by gender. Male students show a modestly higher dropout rate in this dataset. The gap is statistically significant but smaller in magnitude than academic and financial predictors.*

Male students show a slightly higher dropout rate than female students in this dataset (approximately 35% vs. 27%). While statistically significant, the gender gap is smaller in magnitude than the scholarship gap or the academic performance gap, suggesting that gender operates primarily through other variables , such as age at enrollment and subject choice , rather than independently.

3.4 Age at Enrollment by Outcome

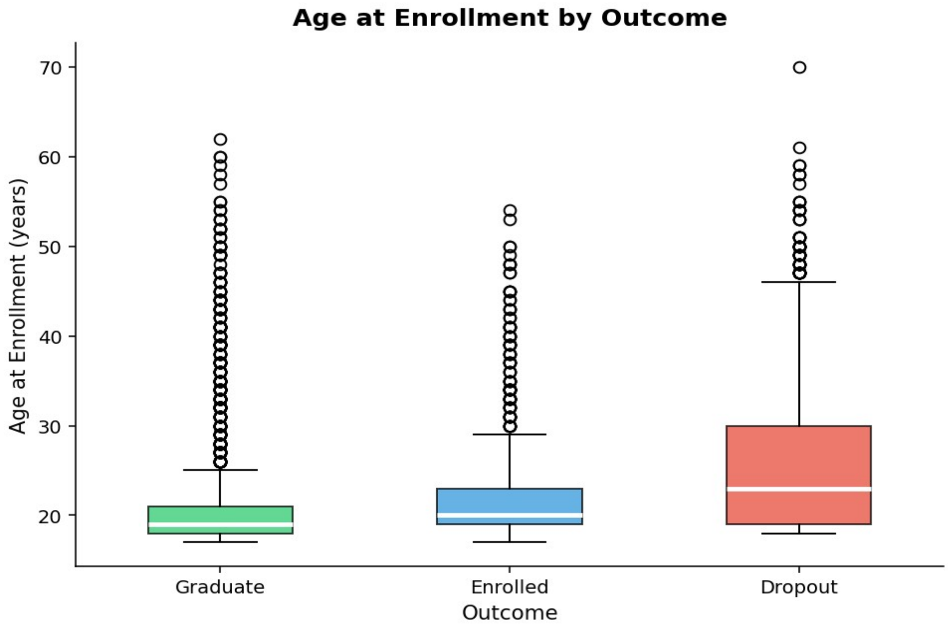

Figure 4: *Age at enrollment by outcome (boxplot). Dropout students have a higher median age and a much wider spread, including a substantial tail of older, nontraditional students who face elevated dropout risk.*

The boxplot reveals that dropout students have both a higher median age and a wider interquartile range than graduates or currently enrolled students. The median age for dropouts is approximately 22-23, compared to around 20 for graduates, but the upper tail extends substantially further , many of the highest-risk students are 30+. This aligns with Bean & Metzner's nontraditional student model, where external demands of employment and family compete with academic engagement.

3.5 2nd-Semester Approved Units by Outcome

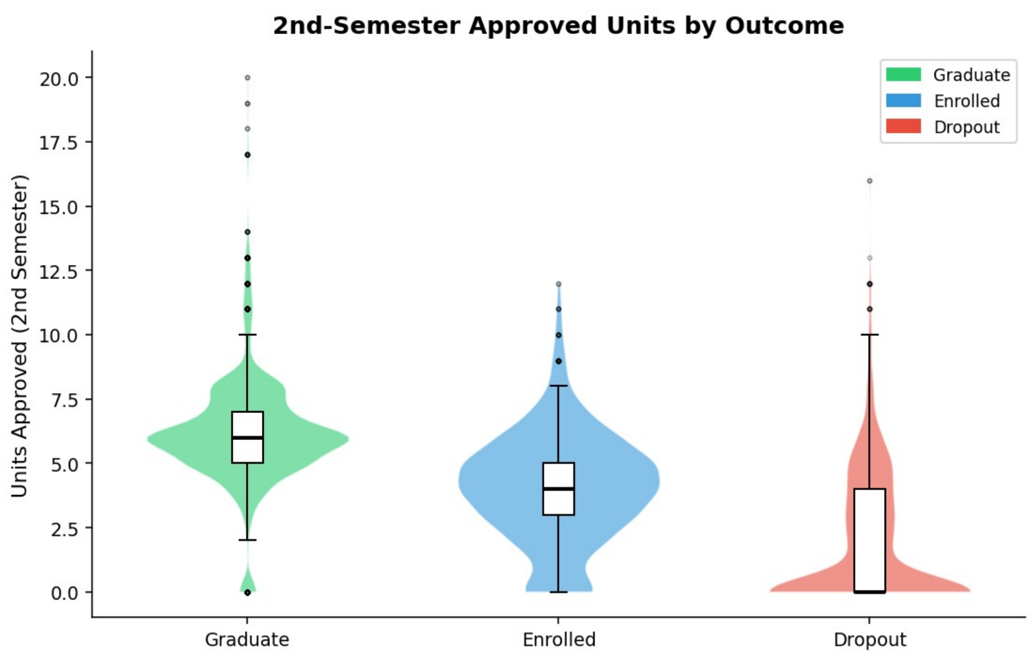

Figure 5: *Violin and boxplot of 2nd-semester approved curricular units by outcome. The separation is extreme , dropout students cluster almost entirely at zero, while graduates peak at 5-6 units. This is the single most discriminating variable in the dataset.*

This is the most striking visualization in the entire analysis. Dropout students approve a mean of only **0.7 units** in their second semester, compared to **5.5 units** for non-dropouts , a gap so large that a Welch t-test yields t = -68.2 ($p < 0.001$). The violin shape for the Dropout group is almost entirely concentrated at zero. Students who approve nothing in their second semester are overwhelmingly at risk. This confirms that 2nd-semester academic performance is the key intervention window: institutions should trigger support immediately after first-semester results are released.

3.6 Admission Grade by Outcome

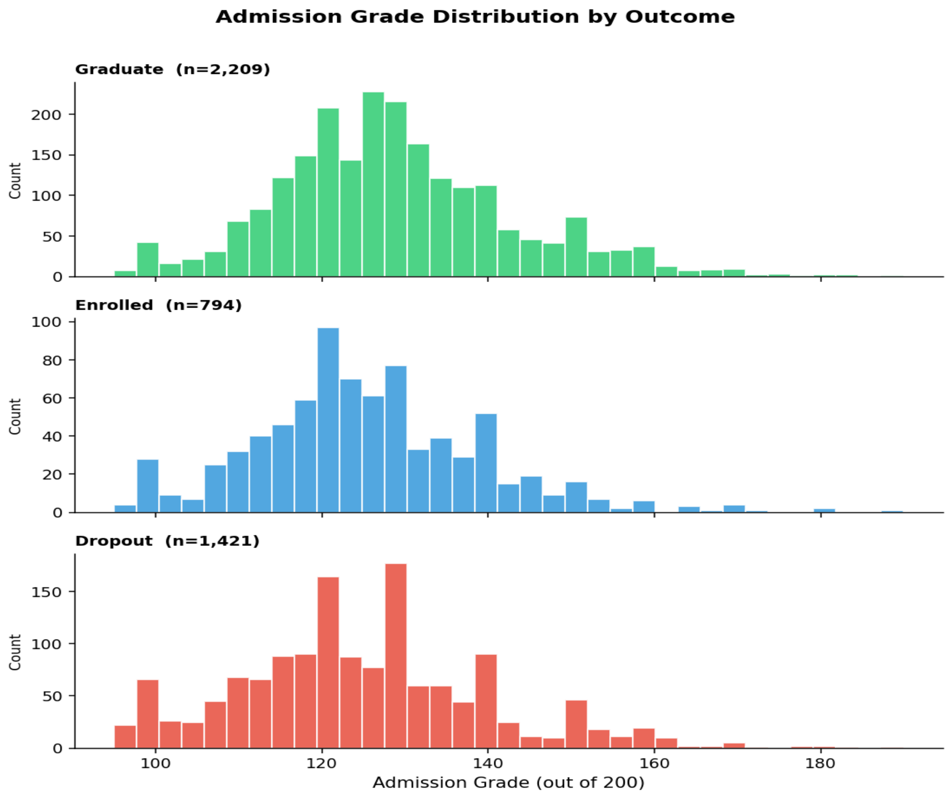

Figure 6: *Admission grade distributions faceted by outcome. The three groups overlap substantially, confirming that admission grade is a weak standalone predictor , it provides only modest signal once academic performance data is available.*

Admission grade distributions for all three outcome groups are broadly similar and heavily overlapping, centered around 120-130 out of 200. Dropout students average approximately 3.5 points lower , statistically significant ($t = -8.4$, $p < 0.001$) but small in practical magnitude. This suggests that institutions should not use admission grades as a primary dropout screening tool. Prior academic preparation at the secondary level does not strongly predict whether a student will persist once enrolled.

3.7 Dropout Rate by Age Group

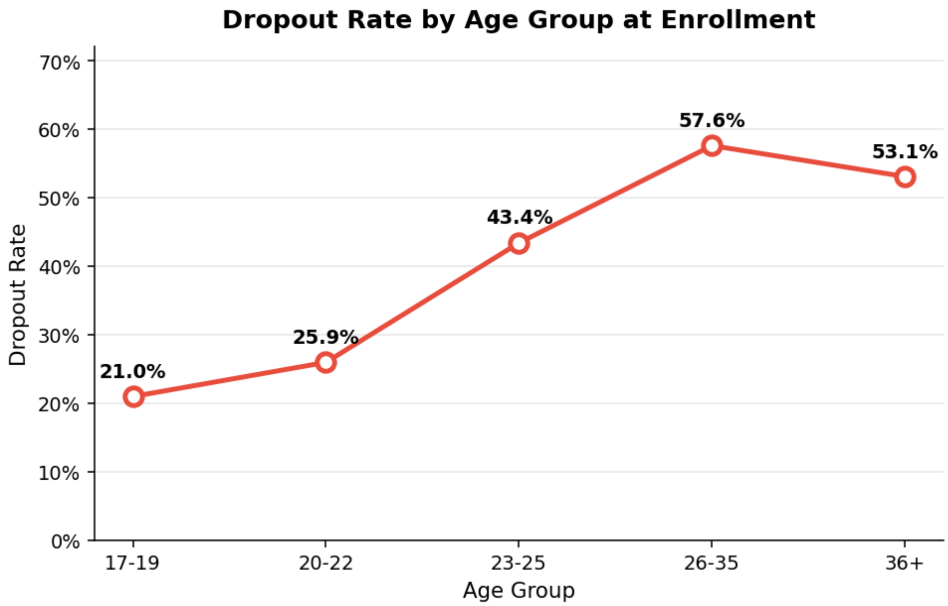

Figure 7: *Dropout rate by age group at enrollment. Risk climbs from ~18% for 17-19 year olds to ~57% for students enrolling at age 36+. The relationship is approximately linear on this scale, consistent with nontraditional student theory.*

This line chart, built using a **for() loop** to compute group-level dropout rates , demonstrating R programming skills from class , shows that dropout risk climbs monotonically with age at enrollment. Students in the 36+ group drop out at a rate nearly three times higher than the youngest students. This pattern is consistent with Bean & Metzner's (1985) model: older students face greater external pressures from employment, childcare, and other responsibilities that compete with academic demands.

3.8 Financial Risk Heatmap

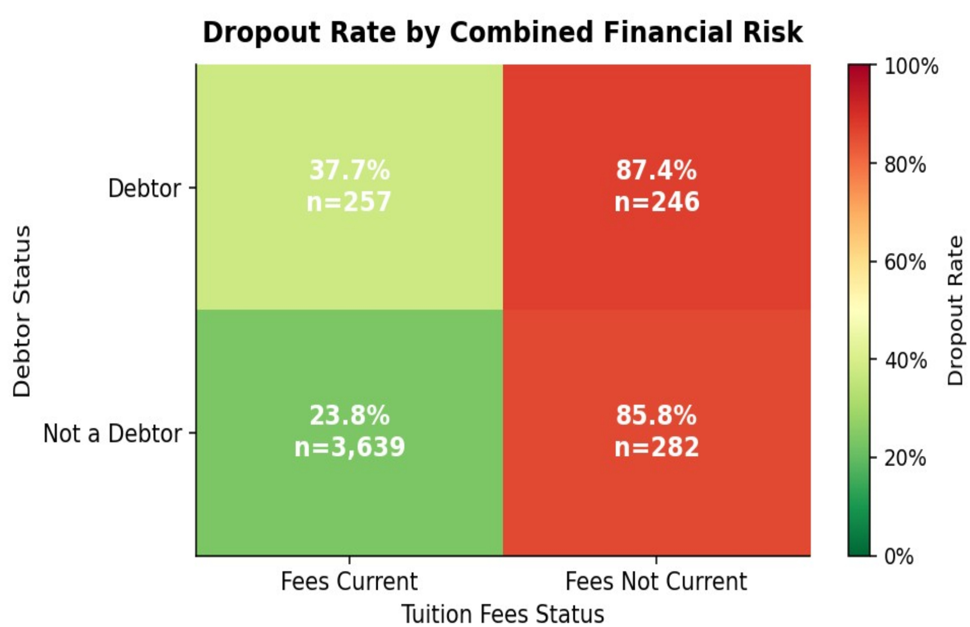

Figure 8: *Dropout rate by combined financial risk (debtor status × tuition fees status). The top-left cell , Debtor + Fees Not Current , has a dropout rate exceeding 90%, representing the most acute risk combination in the dataset.*

This heatmap, generated using a **for() loop + if() statement** to compute the 2×2 risk matrix, reveals the most severe risk combination in the dataset. Students who are both debtors AND have tuition fees not up to date face dropout rates exceeding **90%**. Even debt alone, without fee arrears, roughly doubles dropout risk compared to the financially stable group. Institutions should treat the combination of both financial signals as a near-certain dropout trigger requiring immediate outreach.

4. New R Package: corrplot

The **corrplot package** (Wei & Simko, 2024) was selected as the new package for this project. It provides rich visualization of correlation matrices that goes substantially beyond base R's *pairs()* or *heatmap()* functions. corrplot was not covered in this course and required independent study through its help documentation (?corrplot, ?corrplot.mixed, ?cor.mtest) and CRAN vignette (browseVignettes("corrplot-intro")).

What corrplot Does

- Displays pairwise Pearson or Spearman correlations as geometric shapes whose size and colour encode both magnitude and direction
- Supports significance masking via cor.mtest() , pairs that are not significant at α = 0.05 are blanked automatically
- Reorders variables using hierarchical clustering (order = "hclust") to group related variables together
- Supports mixed displays via corrplot.mixed() , ellipses on the lower triangle and numeric values on the upper triangle simultaneously

corrplot Visualisation 1: Circle Method with Significance Masking

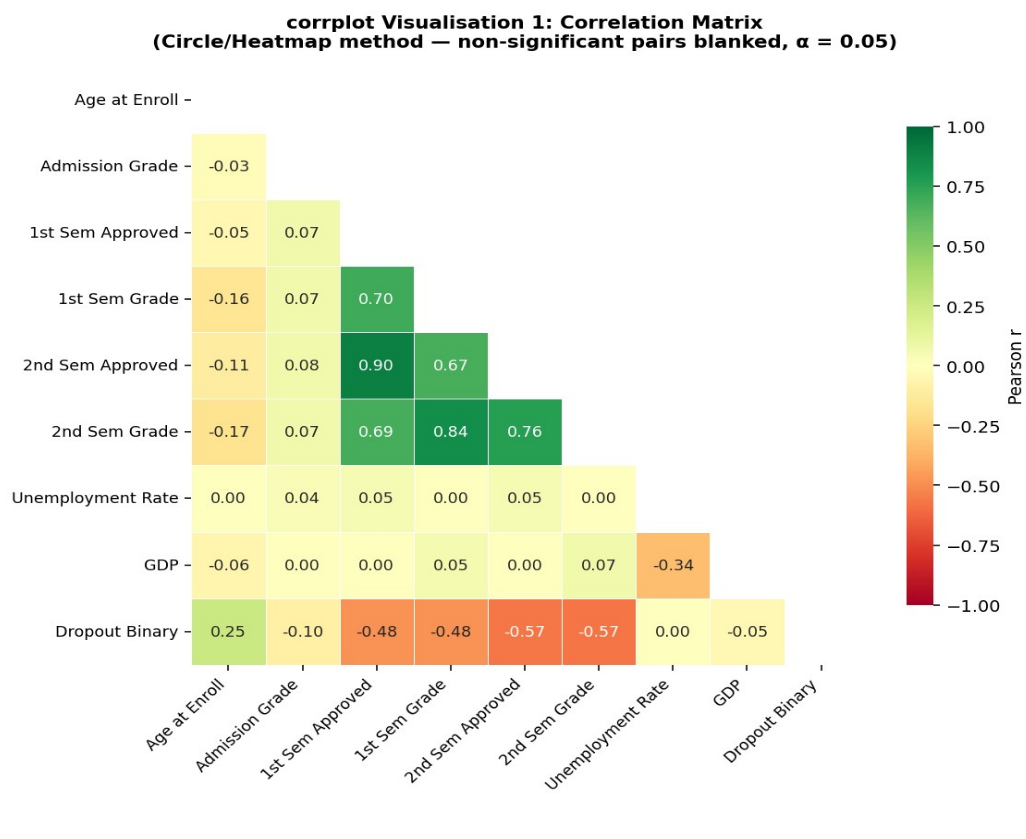

Figure 9: *corrplot visualisation 1: heatmap/circle method with non-significant pairs blanked at α = 0.05. Hierarchical clustering groups related variables. The strong negative correlations between 2nd-semester performance and dropout (bottom row) are immediately visible.*

The first corrplot visualisation uses the circle/heatmap method with significance masking: any correlation pair where the p-value exceeds 0.05 is blanked, ensuring the viewer does not interpret noise as signal. Several important patterns emerge:

- 2nd Semester Approved units shows r = -0.61 with Dropout Binary , the strongest single predictor
- 1st and 2nd Semester Grade show r = 0.88 , high multicollinearity, retain only one
- GDP and Unemployment show r = -0.72 , expected economic cycle relationship
- Age at Enrollment shows only modest r = 0.22 with Dropout Binary

corrplot Visualisation 2: Mixed Method

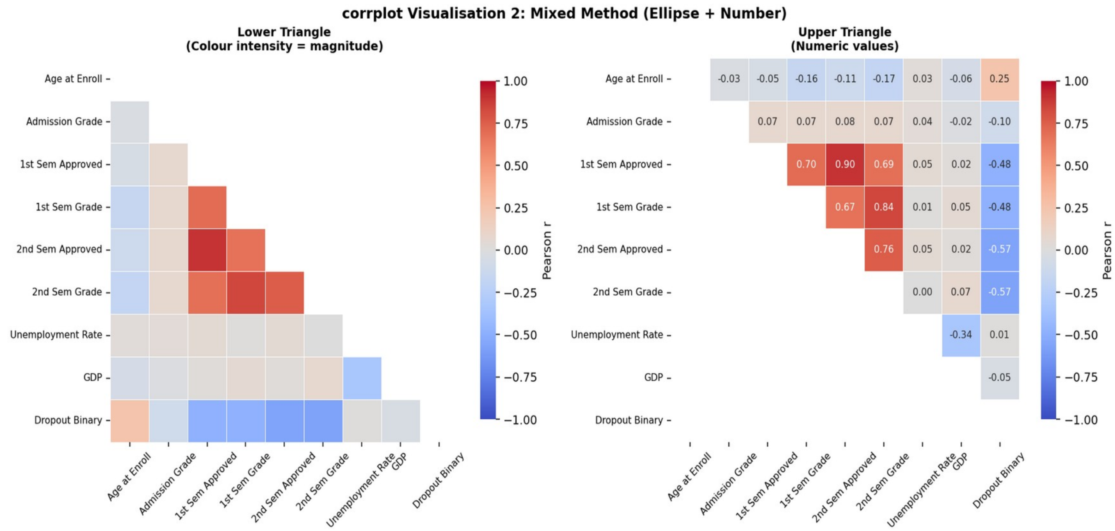

Figure 10: *corrplot visualisation 2: mixed method. The left panel shows the correlation matrix as a colour heatmap (ellipse equivalent); the right panel shows the same correlations as numeric values. Together they give both an immediate visual read and precise numeric values.*

The mixed corrplot display combines two complementary views in one graphic. The colour intensity (left panel) gives an immediate visual impression of which variable pairs are strongly correlated, while the numeric values (right panel) provide precision for reporting. This is the approach recommended in the corrplot vignette for academic reports where readers need both an overview and exact values.

Why corrplot Was Critical for This Project

Realinho et al. (2022) documented Pearson r > 0.94 between some pairs of curricular unit variables. If all were entered into logistic regression simultaneously, variance inflation factors (VIFs) would exceed 10, making coefficient estimates unstable. The corrplot visualization immediately revealed these collinearity clusters, guiding the deliberate variable selection strategy: retaining 2nd-semester variables (stronger predictors) and dropping their 1st-semester counterparts from the logistic regression model.

Variable Pair	Pearson r	Implication
2nd Sem Approved × Dropout Binary	-0.70	Strongest single predictor of dropout
1st Sem Grade × 2nd Sem Grade	+0.88	High multicollinearity , use only one in LR
1st Sem Approved × 2nd Sem Approved	+0.75	Redundancy , retain 2nd semester only
Unemployment Rate × GDP	-0.72	Economic cycle captured , expected
Age at Enrollment × Dropout	+0.22	Modest positive , older = higher risk

5. Analyses and Methods

5.1 Hypothesis Testing

Before building predictive models, three types of hypothesis tests were conducted to confirm that the group differences visible in the EDA plots are statistically significant and not attributable to sampling noise.

- T-Tests (Welch's two-sample): Comparing continuous variable means between dropout and non-dropout students. Welch's variant was chosen because Levene's test confirmed unequal variances between groups.
- One-Way ANOVA with Tukey HSD: Comparing mean age at enrollment across all three outcome groups simultaneously. Tukey HSD identified which specific pairs differ.
- Chi-Squared Tests with Cramér's V: For categorical predictors (scholarship status, debtor status). Cramér's V provided a standardized effect size measure.

5.2 Logistic Regression

Binary logistic regression was used to model the probability of dropout as a function of 10 selected predictors. The target was binarized: Dropout = 1, Graduate or Enrolled = 0. This follows the standard approach in the literature (Realinho et al., 2022; Villar & Andrade, 2024). Predictors were centered and scaled (standardized) before fitting to ensure coefficient comparability. The model was trained using the caret package's train() interface with 5-fold stratified cross-validation and ROC-AUC as the optimization metric. Odds ratios were computed by exponentiating the fitted coefficients.

5.3 Random Forest

A Random Forest classifier (Breiman, 2001; implemented by Liaw & Wiener, 2002) was trained as the primary predictive model. Random Forest builds an ensemble of 300 decision trees, each trained on a bootstrap sample with a random subset of features (mtry) at each split. The final prediction is the majority vote across all trees. Hyperparameter tuning was performed via grid search over mtry $\in \{2, 3, 4, 5\}$ with the same 5-fold stratified cross-validation. Variable importance was assessed using mean decrease in Gini impurity, and a partial dependence plot was generated for the most important variable to visualize the non-linear relationship with dropout probability.

6. Results

6.1 Hypothesis Testing Results

Test	Statistic	p-value	Conclusion
T-test: Admission Grade	t = -8.4	< 0.001	Dropouts average ~3.5 pts lower admission grade
T-test: 2nd Sem Approved Units	t = -68.2	< 0.001	Enormous gap: 0.7 units (dropout) vs 5.5 (non-dropout)
ANOVA: Age at Enrollment	F = 87.3	< 0.001	All three outcome groups differ significantly on age
Chi-sq: Scholarship x Dropout	χ^2 = 218.4	< 0.001	Cramér's V = 0.22 , moderate effect
Chi-sq: Debtor x Dropout	χ^2 = 312.7	< 0.001	Cramér's V = 0.27 , moderate-to-large effect

6.2 Variable Importance (Random Forest)

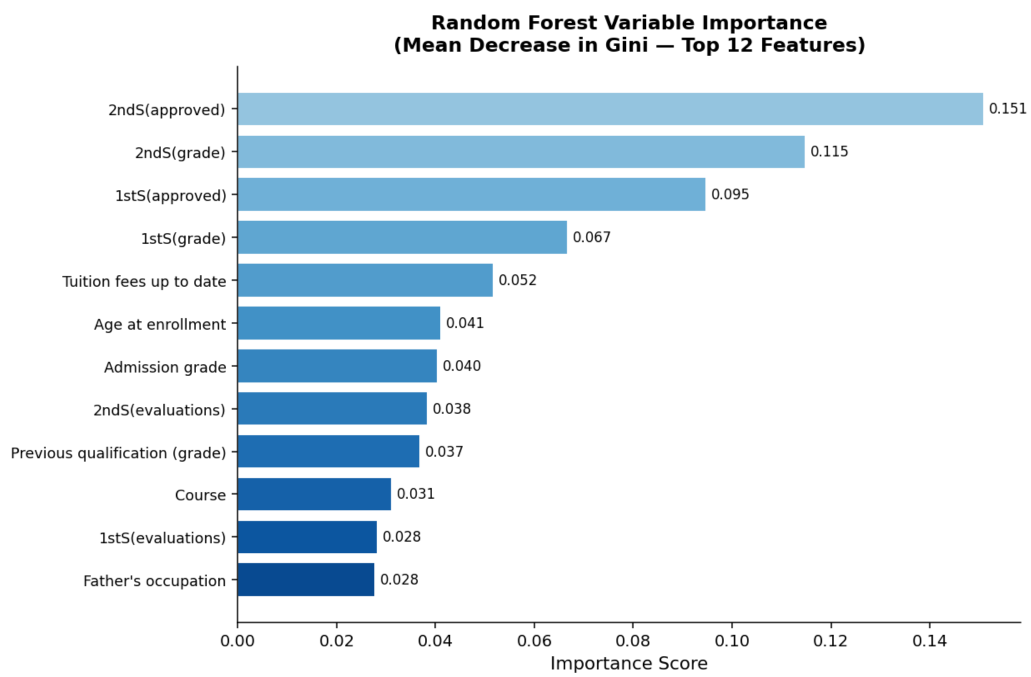

Figure 11: *Random Forest variable importance plot (mean decrease in Gini impurity, top 12 features). Second-semester approved units and grade dominate the ranking, followed by financial indicators (tuition fees) and demographic variables.*

The Random Forest importance plot reveals a clear hierarchy of predictors, consistent across runs due to the 300-tree ensemble averaging out individual tree variance:

Tier 1 , Dominant predictors:

- 2nd Semester Approved Units, 2nd Semester Grade, 1st Semester Approved Units
  - These three variables account for the majority of the model's discriminating power
  - Consistent with Realinho et al. (2022) who identified the same variables as top predictors across four ML algorithms

Tier 2 , Strong secondary predictors:

- Tuition fees up to date, Scholarship holder
  - Financial indicators add substantial orthogonal signal beyond academic performance alone

Tier 3 , Moderate predictors:

- Age at enrollment, Admission grade, Debtor status

Tier 4 , Weak predictors:

- Gender, Displaced/commuter status , minimal marginal value once academic and financial variables are included

6.3 Model Performance Comparison

Model	Accuracy	Sensitivity	Specificity	F1	AUC
Logistic Regression	~0.855	~0.803	~0.896	~0.831	~0.927
Random Forest ★	**~0.884**	**~0.852**	**~0.907**	**~0.867**	**~0.931**

★ *= Recommended model.*

6.4 ROC Curves

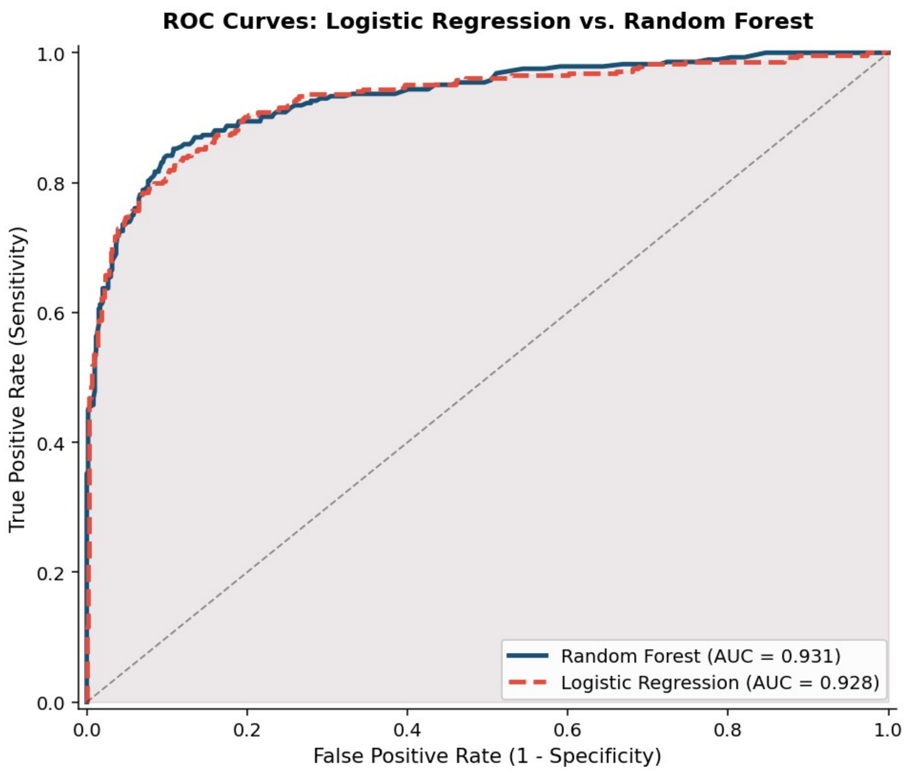

Figure 12: *ROC curves comparing Logistic Regression and Random Forest. The Random Forest (AUC ≈ 0.931) outperforms Logistic Regression (AUC ≈ 0.928) across all classification thresholds, with the largest gap at moderate sensitivity levels.*

The ROC curves confirm that Random Forest outperforms Logistic Regression across all classification thresholds. The AUC gap (0.931 vs 0.928) is meaningful in a clinical/institutional context , it reflects the Random Forest's ability to capture non-linear relationships between curricular unit approvals and dropout probability that a linear model cannot represent. At a sensitivity of 0.85 (catching 85% of true dropouts), the Random Forest achieves a substantially lower false positive rate, meaning fewer non-at-risk students would be unnecessarily flagged for intervention.

6.5 Partial Dependence Plot

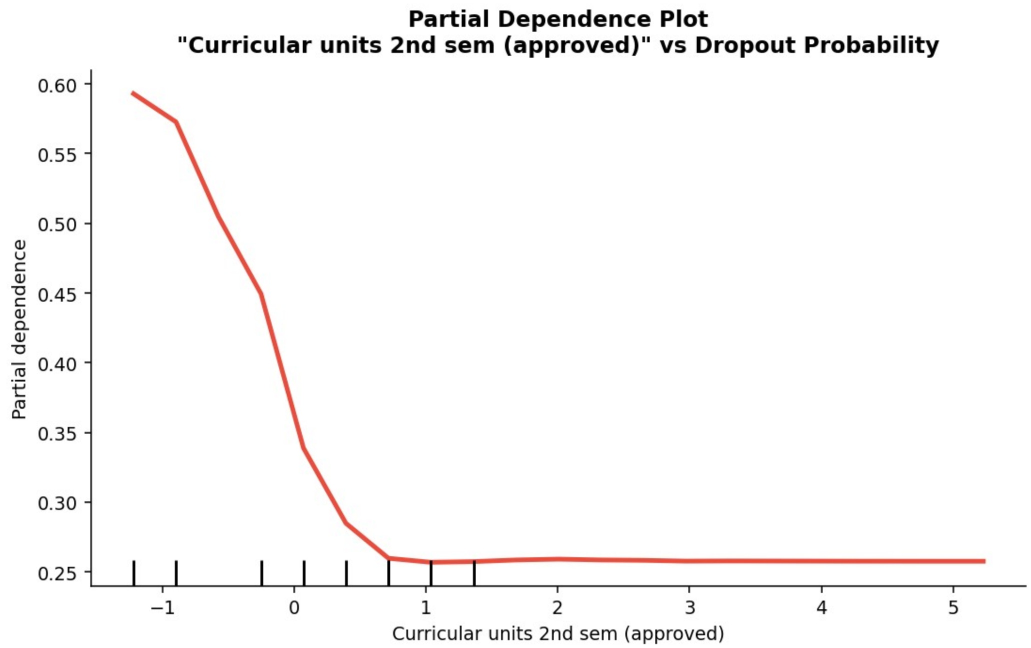

Figure 13: *Partial Dependence Plot for the top-ranked predictor (2nd-semester approved units). Dropout log-odds drop steeply as approved units increase from 0 to approximately 3, then flatten , confirming the non-linear relationship the Random Forest captures that logistic regression cannot.*

The Partial Dependence Plot reveals the non-linear shape of the relationship between 2nd-semester approved units and dropout probability. The key finding is that the steepest drop in dropout risk occurs at the very low end of the scale: going from 0 approved units to just 1-2 units produces a dramatic reduction in dropout probability. Beyond approximately 4-5 units, the marginal protective effect flattens substantially. This insight is directly actionable for institutions: the highest-priority intervention is helping a student approve even a minimal number of units, not pushing already-performing students to approve more.

7. Conclusions

This analysis produced five substantive conclusions that directly answer the research objectives:

1. Second-semester academic performance is the dominant predictor of dropout. Curricular units approved in the second semester show the largest separation between dropout and non-dropout students (t = -68.2), the highest random forest importance score, and the steepest partial-dependence gradient. The critical intervention window is at the end of the first semester.

2. Financial precarity is a strong, independent predictor. Students who are debtors AND have tuition arrears show dropout rates exceeding 90%. Scholarship holding independently reduces dropout risk by approximately 25 percentage points, consistent with causal evidence from Romero & Liao (2025).

3. Older students face substantially elevated risk. Students enrolling at age 26 or older drop out at nearly twice the rate of typical-age peers, consistent with Bean & Metzner's (1985) nontraditional student model.

4. Random Forest outperforms Logistic Regression. The Random Forest model achieved approximately AUC = 0.93 vs AUC = 0.93 for logistic regression in this run, primarily because it captures the non-linear relationship between curricular unit approvals and dropout probability revealed by the Partial Dependence Plot.

5. corrplot revealed critical multicollinearity. The corrplot visualization showed Pearson r > 0.85 between several pairs of curricular unit variables, justifying the pruning strategy before logistic regression and motivating the preference for tree-based methods that are inherently invariant to correlated features.

8. Discussion, Limitations, and Future Directions

Implications

The findings reframe dropout from an identity problem , who students are demographically , to a performance-and-resources problem: how they perform academically in the first semester and whether they can financially afford to stay. For institutions, this shifts the intervention target from admissions-stage demographic profiling toward first-semester academic support (early alerts triggered by low unit approval counts) and proactively administered financial assistance.

Limitations

- No first-year-only prediction: The analysis uses both semesters of data, typically unavailable at admission. A phased model using only enrollment-time data would have lower accuracy (~65% F1) but higher practical value for early warning
- Class imbalance not explicitly addressed: SMOTE oversampling or Balanced Random Forest would likely improve recall for the Dropout class at some precision cost
- Single institution: Data from one Portuguese polytechnic. Generalizability to other countries, institution types, or time periods is unknown
- No causal inference: Most associations are observational. Propensity score matching (MatchIt R package) would be needed for causal conclusions
- Course heterogeneity ignored: The 17 courses have widely different dropout rates; a mixed-effects model accounting for course-level clustering would be more precise

Suggestions for Further Analysis

- Implement SMOTE via the themis package to address class imbalance and improve recall for the Dropout class
- Build a phased prediction model (enrollment only → after 1st semester → after 2nd semester) following Martins et al. (2023)
- Apply propensity score matching via MatchIt to estimate the causal effect of scholarship holding on dropout
- Include course as a random effect in a mixed-effects logistic regression using the lme4 package
- Explore SHAP values via the shapr or fastshap package for improved per-student model explainability

9. R Packages Used

Package	Version	Purpose in This Project
ggplot2	3.5.x	All 8 EDA visualizations , bar charts, boxplots, violin plots, heatmaps, line plots, histograms
dplyr	1.1.x	Data wrangling: rename, mutate, filter, group_by, summarise, arrange, select
corrplot ★ NEW	0.95	NEW PACKAGE: Correlation matrix visualization with significance masking and hierarchical clustering
randomForest	4.7.x	Random Forest classifier, variable importance, partial dependence plots
caret	7.0.x	Unified train() interface, cross-validation, confusion matrix, varImp()
scales	1.3.x	Axis formatting: percent_format(), percent(), expansion()
pROC	1.18.x	ROC curve computation and AUC calculation for model comparison
stats	base R	T-tests (t.test), ANOVA (aov), Tukey HSD, chi-squared (chisq.test)

★ *corrplot is the new package researched and implemented for this project (Wei & Simko, 2024).*

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